

Le Duy Kha

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I define myself as a Full-stack Engineer with 4 years of experience building end-to-end AI and data applications. Expertise spans full-stack development, data pipeline architecture, DevOps automation, and AI system integration. Aspiring Solution Architect focused on delivering innovative AI-driven products and data-intensive applications.

Work Experience:

TechX Company | HCM City
Data Engineer

Dec 2024 - Present

Synaptix AIOps Platform - Incident Detection for Banking Operations

Key Responsibilities:

- Designed and implemented data ingestion pipelines and CI/CD workflows for platform deployment
- Built infrastructure automation and application deployment flows using Helm charts with ArgoCD for automated EKS deployments
- Developed backfill logic and configured networking architecture for production operations
- Challenges: 2MB max size, data ingestion strategy, infra/network management.

Technologies: Helm, EKS, EMR on EKS, Terraform, ArgoCD, GitOps, Temporal

OLAP Reporting System - State Bank of Vietnam

Key Responsibilities:

- Built automated reporting platform for Bank to streamline regulatory and operational reporting
- Developed a complete data pipeline (DBT, Spark, S3 table) on EMR EC2 clusters; faced multi-component setup challenges due to lack of official documentation.

Technologies: DBT, Apache Spark, AWS S3 table (iceberg), EMR on EC2

Xbot - multi agent for operation

Key Responsibilities:

- Designed and proposed the multi-agent architecture, providing technical guidance to the team during the initial development phase
- Developed a scalable automation multi-agent system using LangChain, implementing intelligent routing mechanisms and sub-agent orchestration
- Integrated Qdrant vector database for knowledge retrieval and designed the collection architecture to optimize information access
- Established comprehensive project structure, deployed infrastructure, and implemented CI/CD pipelines for end-to-end service delivery
- Cut infra costs by 60% via EKS to ECS migration.

Technologies: AWS ECS, Terraform, LangChain, Flowise, Langfuse, AWS Services, GitOps

XGenAI - Internal agent chatbot platform

Key Responsibilities:

- Optimized CI/CD workflow, cutting build time and deployment failures by 80%. Defined code merging workflow (failed due to concurrent GitHub agent tag replacement).
- Monitoring system with cost optimization
- Monitored AWS costs, deployed open source saving \$1000/month for company, built Terraform Lambda for auto infrastructure shutdown reducing 50% monthly costs
- Developed backend with SQL API using FastAPI and integrated SQL agent using LLM for querying employee KPI and project allocation data

Technologies Used: Github action, python, fastapi, langgraph, grafana, mirmir, ECR, Cloud trail, Terraform, lambda, bedrock agent

EOD - Enterprise Operation Data for banking

- Build, test, code(Kafka SMT) and config streaming platform for change data capture between multi-SQL and NoSQL database on cloud and on-prem, performance 10k message / min
- Auto Generate config: Build tool for automation generate config for config streaming change capture data from multi-source, reduce 1 hours to config to 1 mins
- Build data quality pipeline
- **Technologies Used:** Kafka, ArgoCD, Docker, AWS, Python, Debezium, Java, K8s, Terragrunt, MongoDB, DocumentDB, Postgres, MySQL, MSSQL, Oracle

Yes4All Company | Phu Nhuan, HCM City

Aug 2023 – Nov 2024

Data Engineer - Data Operation

- **Designed and Implemented Data Pipelines:** Architected, developed, and maintained scalable ETL pipelines to ingest, transform, and load data from diverse sources into a centralized data warehouse, ensuring high data integrity and availability.
- **Collaborated Cross-Functionally:** Worked closely with data scientists, analysts, and other stakeholders to deliver actionable data solutions tailored to business needs.
- **Optimized Data Processing:** Enhanced pipeline performance by 5x through multithreading/multiprocessing techniques and accelerated API data retrieval by 20x using asynchronous I/O.
- **Optimized Database Performance:** Improved database tables and queries using TimescaleDB, partitioning, and indexing methods to ensure efficient data storage and retrieval.
- **Streamlined System Architecture:** Proposed and led the implementation of Kafka for real-time data streaming, significantly improving resource utilization and system throughput.
- **Automated Data Operations:** Developed a bot detection bypass system for web scraping, reducing manual intervention by automating cookie generation for data crawling.
- **Implemented Automation Tools:** Designed and deployed an automation tool for dropshipping bookings, cutting down processing time and reducing manual efforts for the Sales team.
- **Technologies Used:** Python, PostgreSQL, TimescaleDB, Minio, Apache Airflow, Apache Iceberg, Apache Kafka, Google Cloud Service.

Namiq Company | Tan Binh, HCM City

May 2022 – Apr 2023

Python Engineer

- **Developed GraphQL Data Tools:** Researched, designed, and implemented a GraphQL-based database tool, creating efficient CRUD APIs to facilitate seamless data interactions for the Housee mobile application.
- **Implemented Speech AI Models:** Designed and deployed Convolutional Neural Network (CNN) models using mel-spectrogram features for accurate speech classification, enhancing voice-driven data processing.
- **Built NLP Models for Intent Classification:** Collected and preprocessed large datasets, implemented BERT-based models, and deployed them using TorchServe for prediction serving.
- **Designed Robust Data Schemas:** Created optimized database schemas and implemented CRUD APIs with comprehensive unit testing for the Vilik livestream web application, ensuring reliable data management.
- **Technologies Used:** Unity, C#, TypeScript, Hasura, TorchServe, Python, PyTorch, Torch Audio, Flask, PostgreSQL, SQLAlchemy.

Freelance projects:

Nextlab Company | POC Project

Feb 2026

Full-stack Engineer

AI-Powered HR Document Audit Tool

- Clarified Business Requirements Document (BRD), designed solution architecture (LakeHouse Bronze/Silver/Gold, 4-level audit framework), and authored technical specs + API contracts.
- Built AI pipeline (Gemini 2.0 Flash) for classifying 32 document types, quality checks, and OCR extraction.
- Developed a 3-level audit engine, Next.js dashboard, and Excel reports.
- Delivered a full-stack Proof of Concept (POC) in ~15 hours.
- **Technologies Used:** Python, FastAPI, Next.js, TypeScript, Google Gemini 2.0 Flash, Terraform, Cloud Run, Docker.

Mernva technologies Company | Remote part time - India

Jul 2023 – Nov 2024

As a Platform Engineer, I integrate AI-driven data processing models and pipelines into backend systems, collaborating remotely with international teams to enhance data engineering processes.

- **Integrated AI Models into Data Pipelines:** Seamlessly incorporated AI-driven models into backend systems, enhancing data processing capabilities and enabling intelligent data transformations.
- **Chat Bot Document Summary | LLM Model:** Conducted research and development on a chatbot model using Retrieval-Augmented Generation (RAG) and APIs for efficient data retrieval and summary generation.
- **Technologies Used:** AWS Cloud (EC2, K8s), Python, FastAPI, Docker, Kubernetes, LangChain, Hugging Face.

Education:

Bachelor degree | VNUHCM – University of science | Computer science - GPA: 8.4 / 10

High school | LUONG THE VINH HIGH SCHOOL FOR THE GIFTED | 2016 – 2019

Skills:

- Soft skills: Teamwork, communication, listening, leadership.
- **Technical skill:**
 - **Programming Languages:** Python, C++, Rust, Golang, Type script
 - **Frameworks:** PySpark, Selenium, FastAPI, PyTorch, React
 - **Databases:** PostgreSQL, Redis, MySQL, MongoDB, MinIO, Iceberg
 - **Skills:** Git, Gitlab, Linux, Bash Script, AWS Cloud, Kubernetes
 - **Language:** English (TOEIC 765), VietNameese (Native)

Certifications:

- [AWS Certified Solutions Architect – Associate](#) 2025
- Top 2 in MedTech domain, HackYouth 2023 Vin Uni, 2023
- Machine learning in production, online course(coursera), 2022
- SAT Math level 2 (800 / 800), SAT exam, 2021
- Deep learning specialization, online course(coursera), 2021

EXTRA-CURRICULAR ACTIVITIES:

Charity Work | Volunteer at [HOPE](#) | 2024 - Now

[MLOpsVN](#) | Coordinator - Community for Machine Learning Engineer | 2023

[SAB in HCMUS](#) | Head of department - Student Organization | 2020 – 2022